

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

<thought>

To solve this problem , we need to identify the artery that is in the anatomic snuff box , which is close to the thumb , and find out which one is most likely damaged given the scenario .

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<answer>

A . Radial artery </answer>

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The anatomical snuffbox area includes the thumb (i 5), index (i 2), middle (i 3), and ring (i 4) fingers , and it is bordered by the tend ons of the deep flex or pol lic is long us and bre vis . The main arterial supply for this area is the superficial pal mar arch , which passes through the snuff box and divides into the terminal branches of the superficial pal mar arch : the recurrent and superficial arch of the ul nar artery and the terminal branches : the superficial pal mar terminal branches of the radial artery . As the injury happened due to a penetrating knife stab , it is the likelihood that the **radial artery** is the most likely injured artery causing excessive bleeding .

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